

Initiating Search

November 4, 2025, 11:50 AM

• Search:

CAS Lexicon Search:

Miyaura borylation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (284)	 References	View Results
Exported: Retrieved Related Reaction Results (1,057)	 Reactions	View Results

Copyright © 2025 American Chemical Society (ACS). All Rights Reserved.

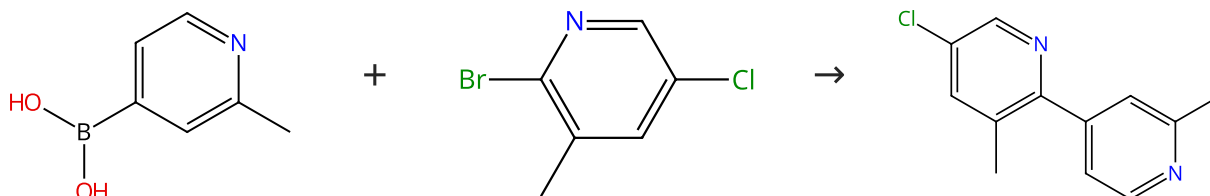
Internal use only. Redistribution is subject to the terms of your CAS SciFinder License Agreement and CAS information Use Policies.

Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (83)

 Suppliers (74)

 Supplier (1)

31-179-CAS-18355942

Steps: 1 Yield: 100%

Use of reaction medium containing water-surfactant mixture in cross-coupling and other reactions

Assignee: Novartis AG

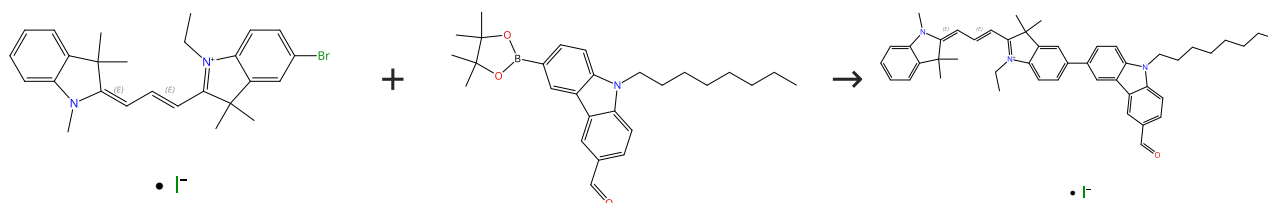
World Intellectual Property Organization, WO2017168303 A1 2017-10-05

[PatentPak](#) available

1.1 **Reagents:** Triethylamine, Poly(oxy-1,2-ethanediyl), α-[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]-ω-methoxy-
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Tetrahydrofuran, Water; 18 h, 40 °C

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

31-614-CAS-34696734

Steps: 1 Yield: 100%

Synthesis of Novel Key Chromophoric Intermediates via C-C Coupling Reactions

By: Nhari, Laila M.; et al

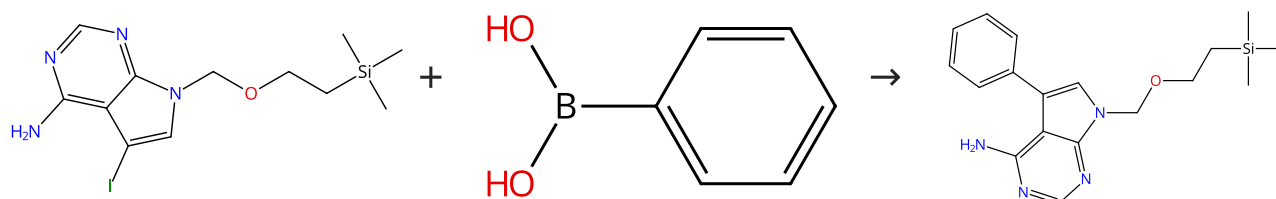
Catalysts (2022), 12(10), 1292.

1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Methanol, Toluene; 30 min, 80 °C

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 99%

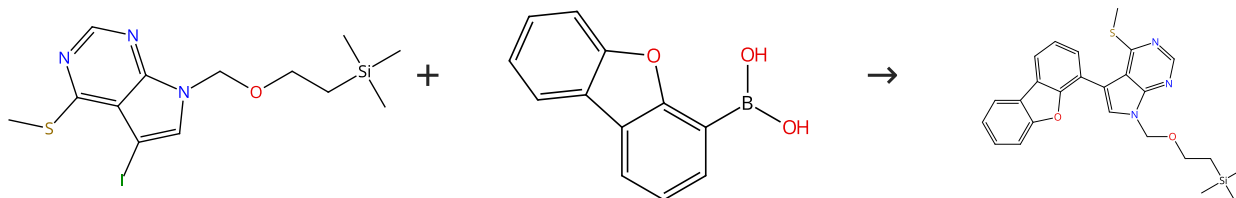

 Suppliers (12)

 Suppliers (127)

31-179-CAS-18370395 Steps: 1 Yield: 99% 1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3) Solvents: Acetonitrile, Water; 3 h, 100 °C Experimental Protocols	Synthesis of 2,6-Substituted 7-(Het)aryl-7-deazapurine Nucleobases (2,4-Disubstituted 5-(Het)aryl-pyrrolo[2,3-d]pyrimidines) By: Sabat, Nazarii; et al Synthesis (2017), 49(20), 4623-4650.
--	---

Scheme 4 (1 Reaction)

Steps: 1 Yield: 99%

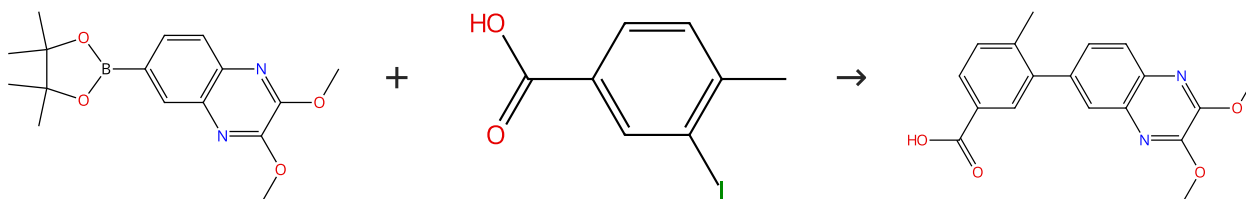


Suppliers (92)

31-179-CAS-18370414 Steps: 1 Yield: 99% 1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3) Solvents: Acetonitrile, Water; 3 h, 100 °C Experimental Protocols	Synthesis of 2,6-Substituted 7-(Het)aryl-7-deazapurine Nucleobases (2,4-Disubstituted 5-(Het)aryl-pyrrolo[2,3-d]pyrimidines) By: Sabat, Nazarii; et al Synthesis (2017), 49(20), 4623-4650.
--	---

Scheme 5 (1 Reaction)

Steps: 1 Yield: 99%

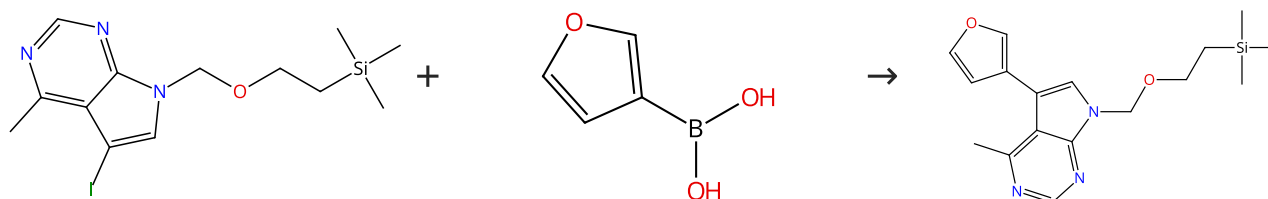


Suppliers (90)

31-179-CAS-18346063 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 4 h, 90 °C	Revisiting the Quinoxalinedione Scaffold in the Construction of New Ligands for the Ionotropic Glutamate Receptors By: Demmer, Charles S.; et al ACS Chemical Neuroscience (2017), 8(11), 2477-2495.
--	--

Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%

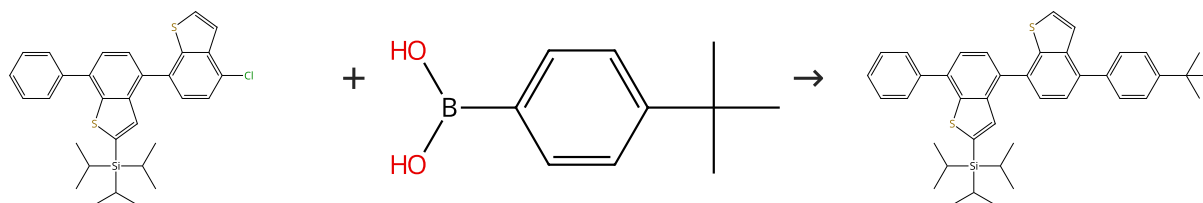


Suppliers (101)

31-179-CAS-18370416 Steps: 1 Yield: 99% 1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3) Solvents: Acetonitrile, Water; 1 h, 100 °C Experimental Protocols	Synthesis of 2,6-Substituted 7-(Het)aryl-7-deazapurine Nucleobases (2,4-Disubstituted 5-(Het)aryl-pyrrolo[2,3-d]pyrimidines) By: Sabat, Nazarii; et al Synthesis (2017), 49(20), 4623-4650.
--	---

Scheme 7 (1 Reaction)

Steps: 1 Yield: 99%

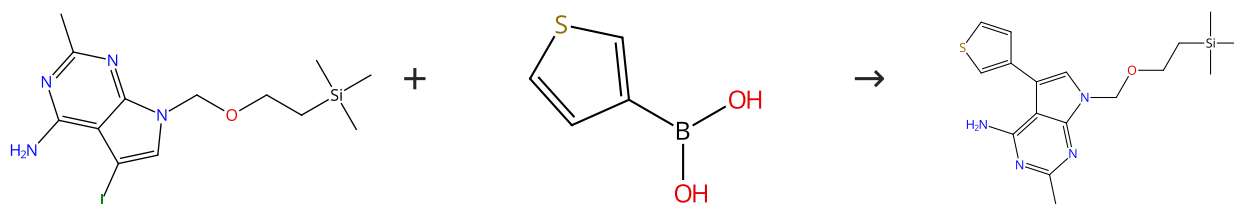


Suppliers (92)

31-614-CAS-24280099 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: (<i>SP</i>4-3)-[2'-(Amino-κ<i>M</i>)[1,1'-biphenyl]-2-yl-κ<i>C</i>][dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine](methanesulfonato-κ<i>O</i>)palladium Solvents: 1,4-Dioxane, Water; 22 h, 90 °C Experimental Protocols	Synthesis of 4,7'-Bibenzo[b]thiophenes Bearing Several Different Substituents at 2-, 2'-, 4'-, and 7-Positions; Structurally Featured Molecular Scaffolds for Selective Substitution By: Mikami, Shinichi; et al Synlett (2021), 32(18), 1826-1832.
--	---

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%

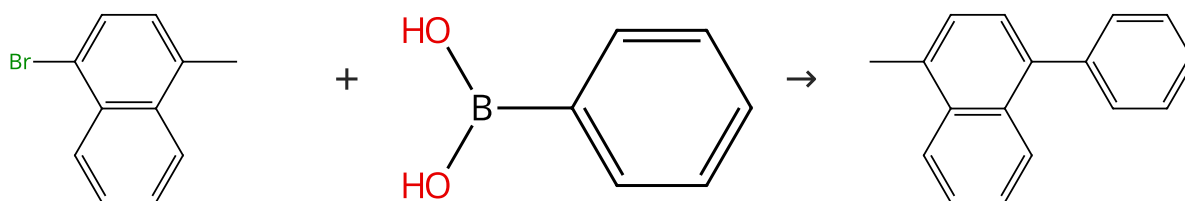


Suppliers (100)

31-179-CAS-18370400 Steps: 1 Yield: 99% 1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3) Solvents: Acetonitrile, Water; 3 h, 100 °C Experimental Protocols	Synthesis of 2,6-Substituted 7-(Het)aryl-7-deazapurine Nucleobases (2,4-Disubstituted 5-(Het)aryl-pyrrolo[2,3-d]pyrimidines) By: Sabat, Nazarii; et al Synthesis (2017), 49(20), 4623-4650.
--	---

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (67)

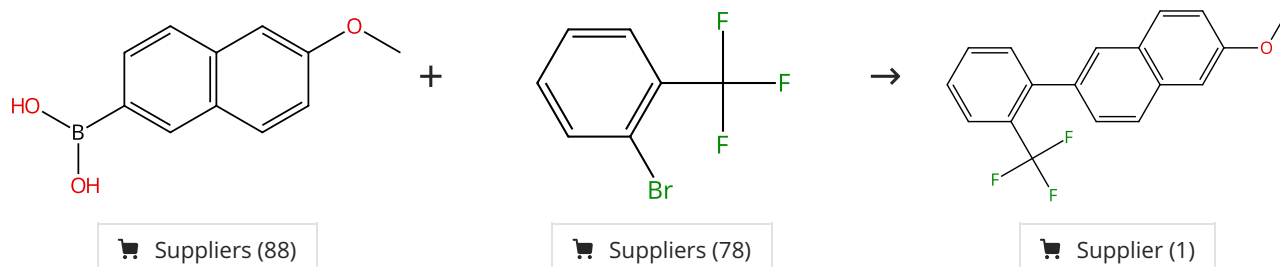
Suppliers (127)

Suppliers (10)

31-179-CAS-3093553	Steps: 1 Yield: 99%	"Nok": A Phytosterol-Based Amphiphile Enabling Transition-Metal-Catalyzed Couplings in Water at Room Temperature
1.1 Reagents: Triethylamine, Poly(oxy-1,2-ethanediyl), α -[1,4-dioxo-4-[(3 β)-stigmast-5-en-3-yloxy]butyl]- ω -methoxy- Catalysts: (SP-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino- κ^2]ferrocene]dichloropalladium Solvents: Water; 2 h, rt		By: Klumphu, Piyatida; et al Journal of Organic Chemistry (2014), 79(3), 888-900.
Experimental Protocols		

Scheme 10 (1 Reaction)

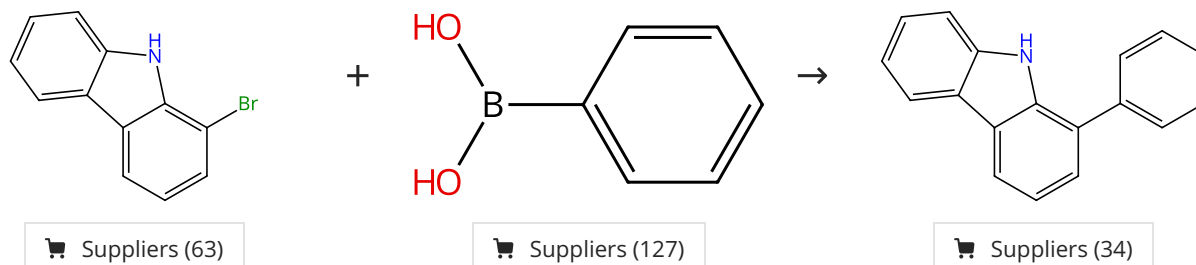
Steps: 1 Yield: 99%



31-179-CAS-7301878	Steps: 1 Yield: 99%	"Nok": A Phytosterol-Based Amphiphile Enabling Transition-Metal-Catalyzed Couplings in Water at Room Temperature
1.1 Reagents: Triethylamine, Poly(oxy-1,2-ethanediyl), α -[1,4-dioxo-4-[(3 β)-stigmast-5-en-3-yloxy]butyl]- ω -methoxy- Catalysts: (SP-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino- κ^2]ferrocene]dichloropalladium Solvents: Water; 4 h, rt		By: Klumphu, Piyatida; et al Journal of Organic Chemistry (2014), 79(3), 888-900.
Experimental Protocols		

Scheme 11 (1 Reaction)

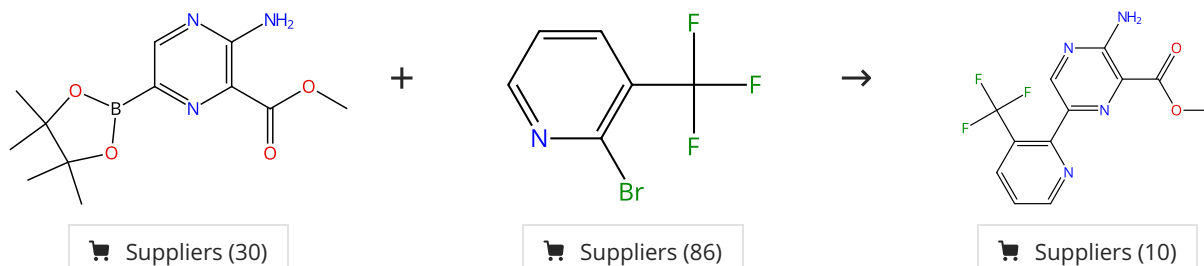
Steps: 1 Yield: 99%



31-614-CAS-39600584	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 12 (1 Reaction)

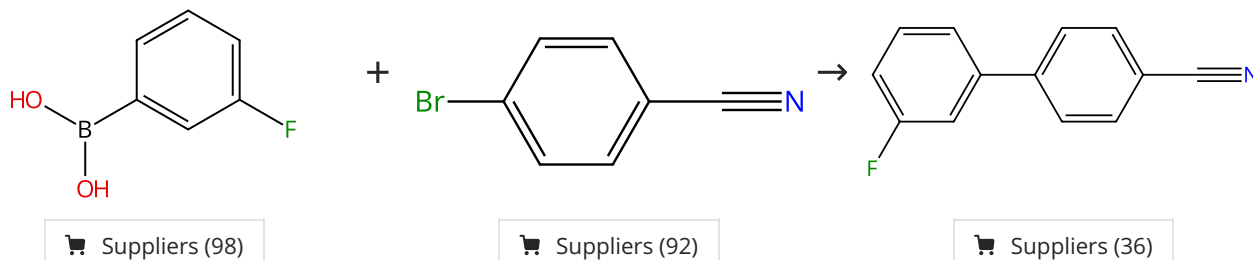
Steps: 1 Yield: 99%



31-179-CAS-18355936 Steps: 1 Yield: 99%	Use of reaction medium containing water-surfactant mixture in cross-coupling and other reactions Assignee: Novartis AG World Intellectual Property Organization, WO2017168303 A1 2017-10-05 PatentPak available
1.1 Reagents: Triethylamine, Poly(oxy-1,2-ethanediyl), α -[4-[[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]- ω -methoxy- Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Acetone, Water; rt; 1 h, rt	

Scheme 13 (1 Reaction)

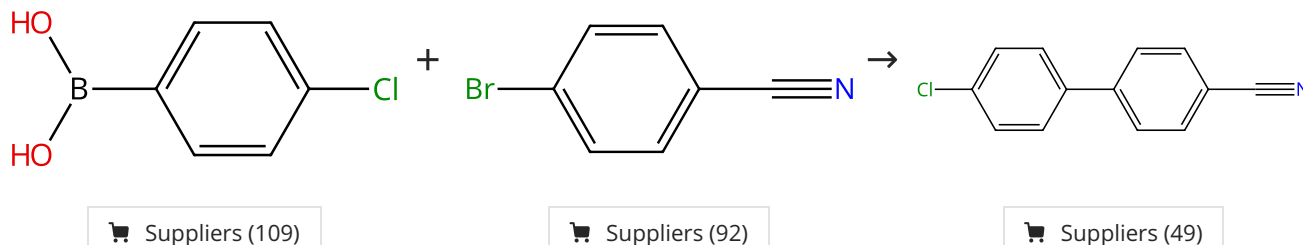
Steps: 1 Yield: 99%



31-614-CAS-39600563 Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C	

Scheme 14 (1 Reaction)

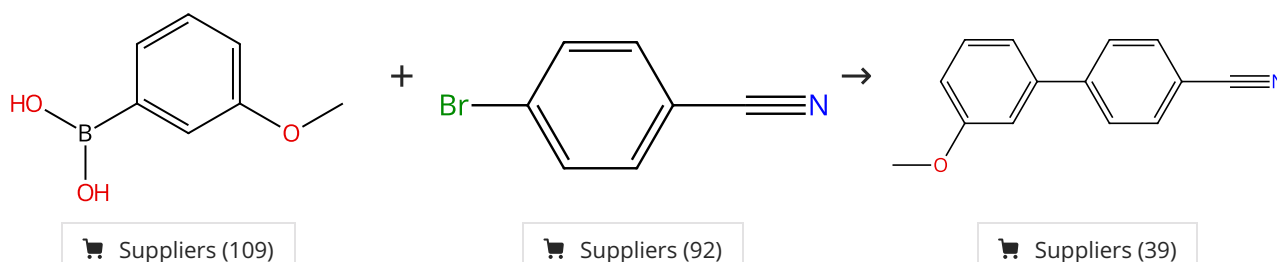
Steps: 1 Yield: 99%



31-614-CAS-39600569 Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C	

Scheme 15 (1 Reaction)

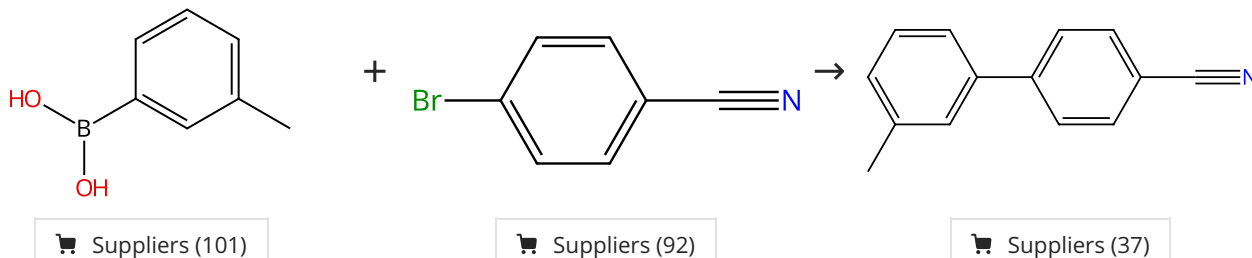
Steps: 1 Yield: 99%



31-614-CAS-39600564	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 16 (1 Reaction)

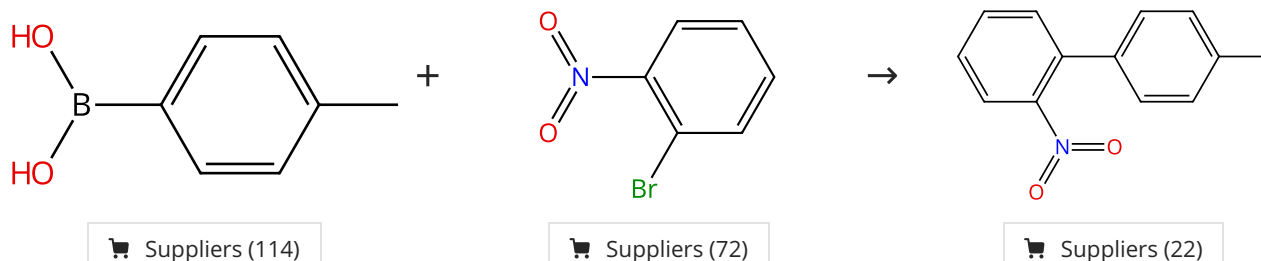
Steps: 1 Yield: 99%



31-614-CAS-39600576	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 17 (1 Reaction)

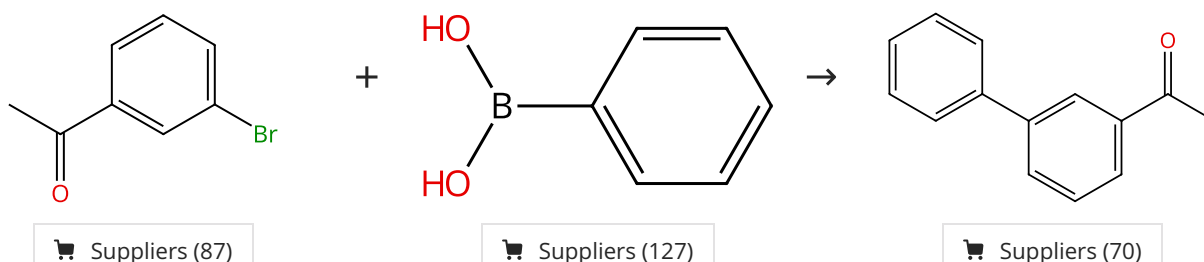
Steps: 1 Yield: 99%



31-614-CAS-39600578	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 18 (1 Reaction)

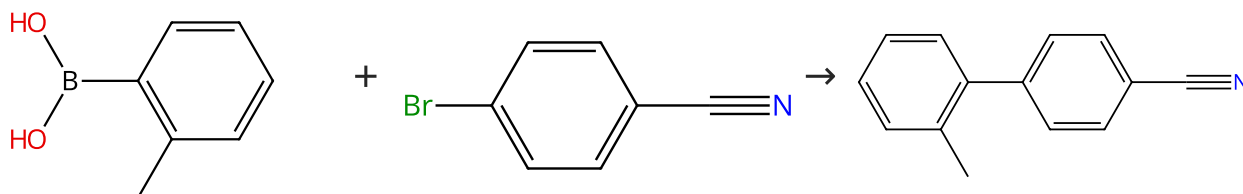
Steps: 1 Yield: 99%



31-614-CAS-39600586	Steps: 1 Yield: 99% Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C	Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (106)

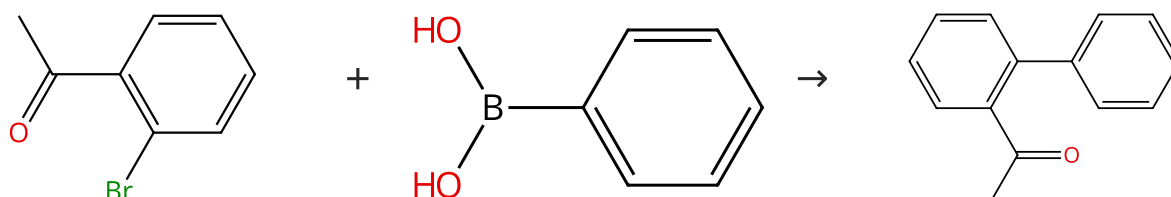
Suppliers (92)

Suppliers (46)

31-614-CAS-39600570	Steps: 1 Yield: 99% Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C	Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (89)

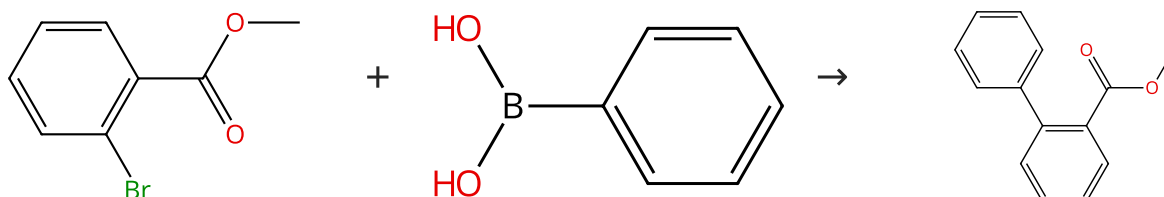
Suppliers (127)

Suppliers (63)

31-614-CAS-39600579	Steps: 1 Yield: 99% Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C	Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 21 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (78)

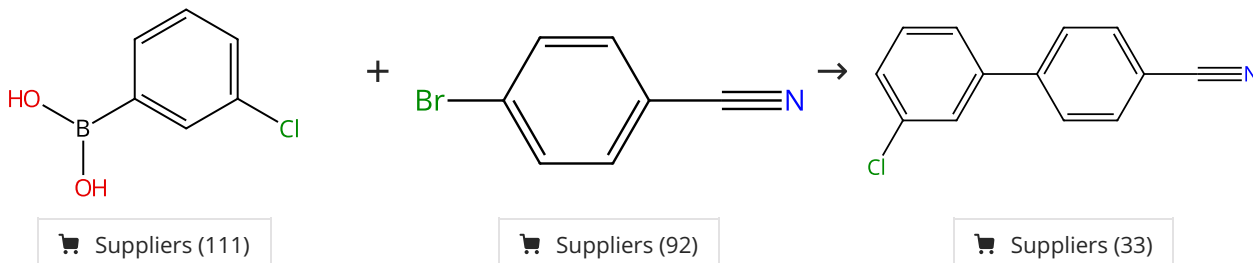
Suppliers (127)

Suppliers (52)

31-614-CAS-39600582	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 22 (1 Reaction)

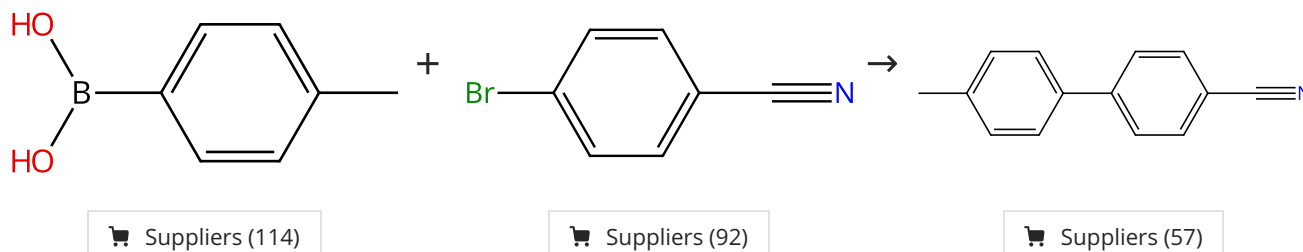
Steps: 1 Yield: 99%



31-614-CAS-39600572	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 23 (1 Reaction)

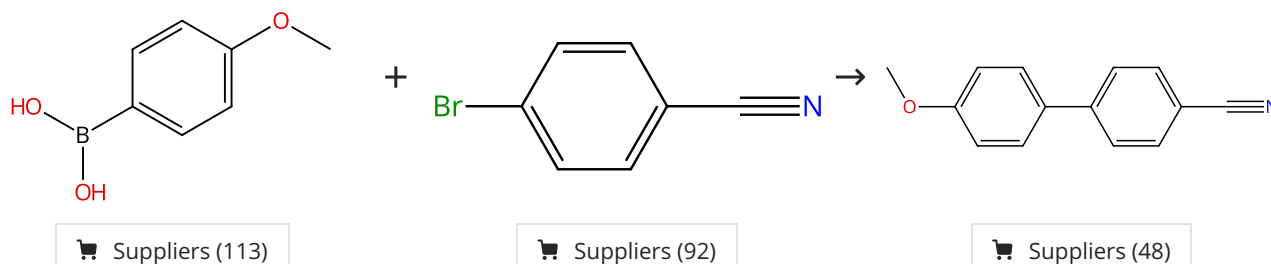
Steps: 1 Yield: 99%



31-614-CAS-39600574	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 24 (1 Reaction)

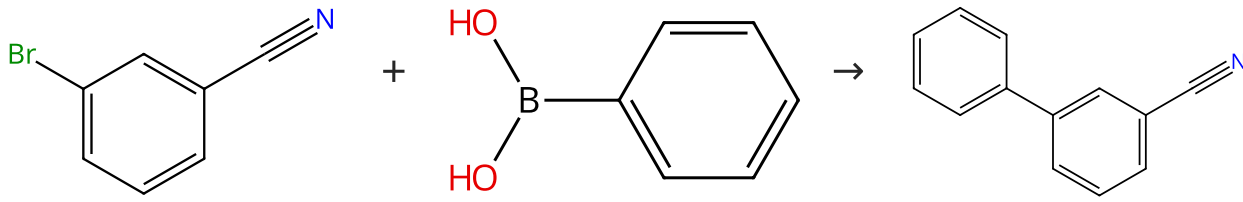
Steps: 1 Yield: 99%



31-614-CAS-39600568	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 25 (3 Reactions)

Steps: 1 Yield: 99%



Reaction scheme showing the synthesis of 1-cyanobiphenyl-4-yl from 4-bromo-1-cyanobenzene and phenylboronic acid.

4-bromo-1-cyanobenzene (with Br highlighted in green and CN highlighted in blue) reacts with phenylboronic acid (with HO groups highlighted in red) to form 1-cyanobiphenyl-4-yl (with CN highlighted in blue).

 Suppliers (79)

 Suppliers (127)

 Suppliers (54)

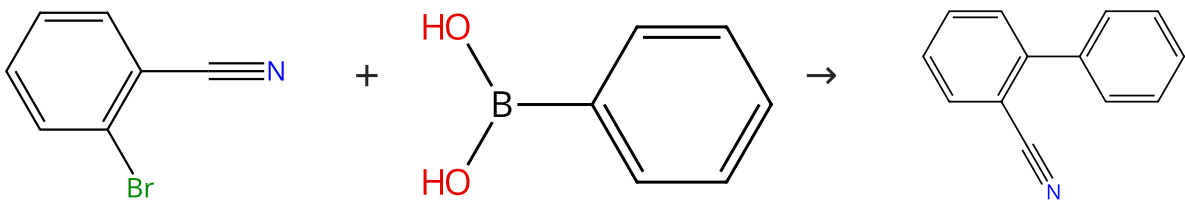
31-614-CAS-39600580	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

31-179-CAS-7046064	Steps: 1 Yield: 99%	"Nok": A Phytosterol-Based Amphiphile Enabling Transition-Metal-Catalyzed Couplings in Water at Room Temperature
1.1 Reagents: Triethylamine, Poly(oxy-1,2-ethanediyl), α-[1,4-dioxo-4-[(3β)-stigmast-5-en-3-yloxy]butyl]-ω-methoxy-Catalysts: (SP-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino-κP]ferrocene]dichloropalladium Solvents: Water; 2 h, rt		By: Klumphu, Piyatida; et al Journal of Organic Chemistry (2014), 79(3), 888-900.
Experimental Protocols		

31-179-CAS-939247	Steps: 1 Yield: 99%	"Nok": A Phytosterol-Based Amphiphile Enabling Transition-Metal-Catalyzed Couplings in Water at Room Temperature
1.1 Reagents: Triethylamine, Poly(oxy-1,2-ethanediyl), α-[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]-ω-methoxy-Catalysts: (SP-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino-κP]ferrocene]dichloropalladium Solvents: Water; 2 h, rt		By: Klumphu, Piyatida; et al Journal of Organic Chemistry (2014), 79(3), 888-900.
Experimental Protocols		

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Chemical reaction scheme showing the synthesis of 1-cyano-2-phenylbenzene. The reaction involves 2-bromo-1-cyanobenzene (left) reacting with phenylboronic acid (middle) to produce 1-cyano-2-phenylbenzene (right). The bromine atom is highlighted in green, and the cyano group is highlighted in blue.

 Suppliers (78)

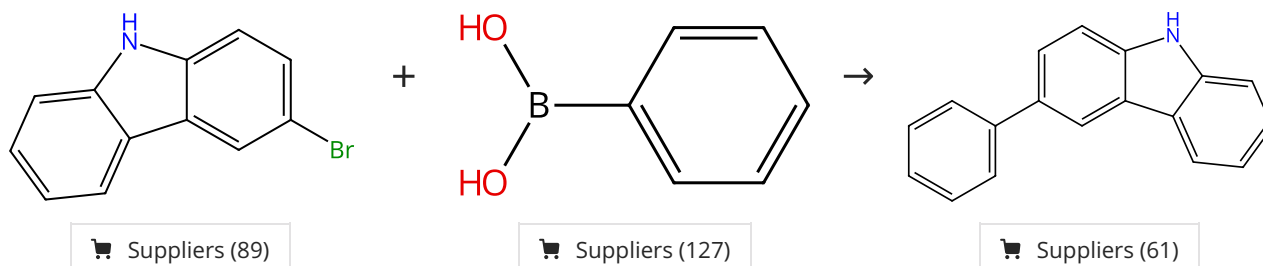
 Suppliers (127)

 Suppliers (61)

31-614-CAS-39600575	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 27 (1 Reaction)

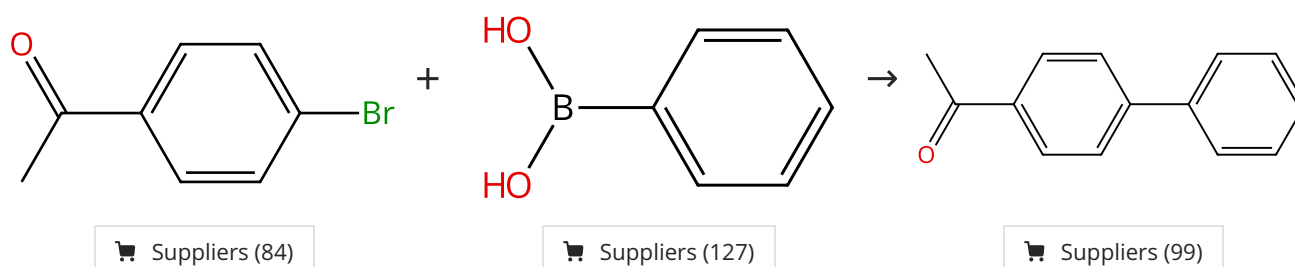
Steps: 1 Yield: 99%



31-614-CAS-39600583	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 28 (1 Reaction)

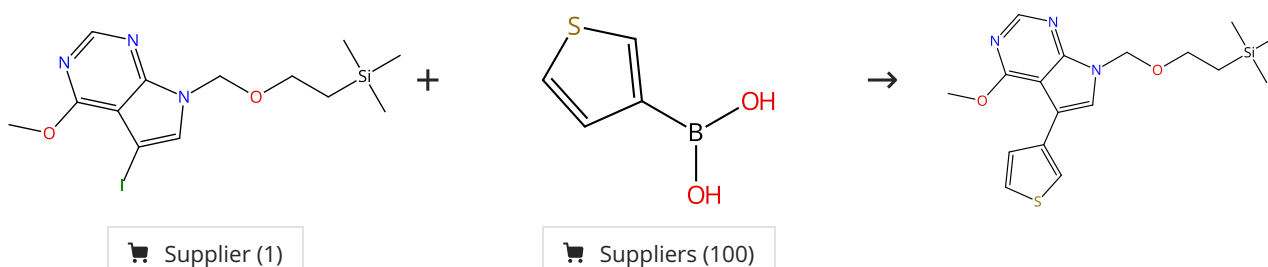
Steps: 1 Yield: 99%



31-614-CAS-39600581	Steps: 1 Yield: 99%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 29 (1 Reaction)

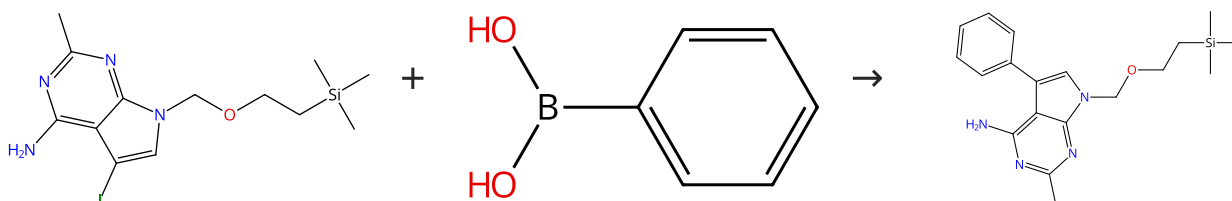
Steps: 1 Yield: 98%



31-179-CAS-18370405	Steps: 1 Yield: 98%	Synthesis of 2,6-Substituted 7-(Het)aryl-7-deazapurine Nucleo bases (2,4-Disubstituted 5-(Het)aryl-pyrrolo[2,3-d]pyrimidines)
1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3) Solvents: Acetonitrile, Water; 3 h, 100 °C		By: Sabat, Nazarii; et al Synthesis (2017), 49(20), 4623-4650.
Experimental Protocols		

Scheme 30 (1 Reaction)

Steps: 1 Yield: 98%

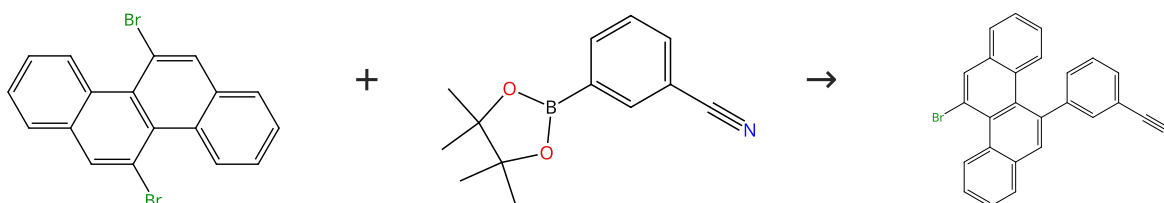


Suppliers (127)

31-179-CAS-18370402	Steps: 1 Yield: 98%	Synthesis of 2,6-Substituted 7-(Het)aryl-7-deazapurine Nucleo bases (2,4-Disubstituted 5-(Het)aryl-pyrrolo[2,3-d]pyrimidines)
1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3) Solvents: Acetonitrile, Water; 3 h, 100 °C		By: Sabat, Nazarii; et al Synthesis (2017), 49(20), 4623-4650.
Experimental Protocols		

Scheme 31 (1 Reaction)

Steps: 1 Yield: 98%



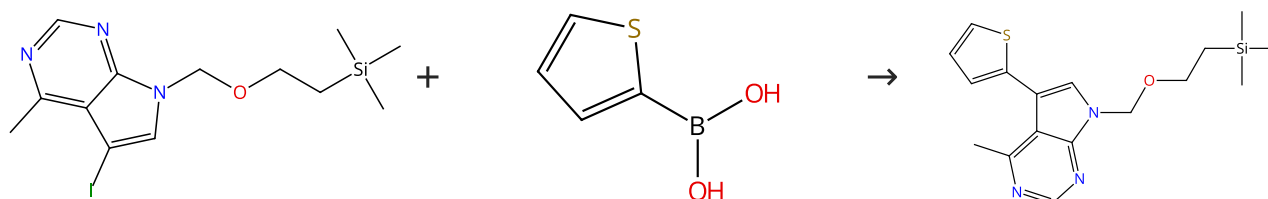
Supplier (1)

Suppliers (76)

31-614-CAS-42793823	Steps: 1 Yield: 98%	Preparation of chrysene derivatives for organic electrolu minescent devices
1.1 Reagents: Potassium carbonate Catalysts: Tetrabutylammonium bromide, Tetrakis(triphenyl phosphine)palladium Solvents: Toluene; rt → 90 °C; 12 h, 90 °C		Assignee: South China Institute of Collaborative Innovation China, CN118026915 A 2024-05-14 PatentPak available

Scheme 32 (1 Reaction)

Steps: 1 Yield: 98%

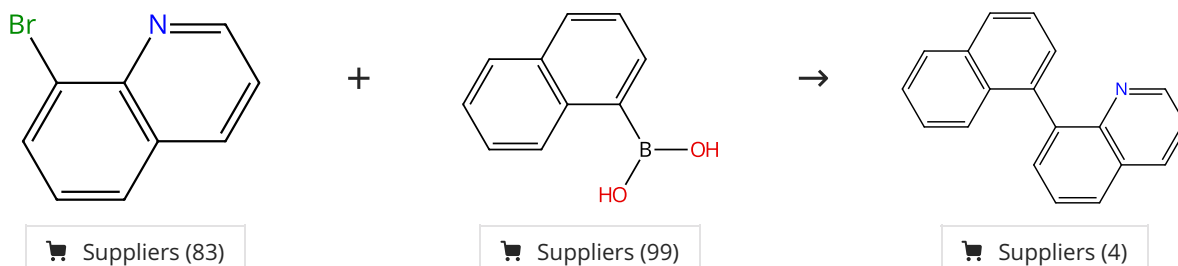


Suppliers (102)

31-179-CAS-18370417	Steps: 1 Yield: 98%	Synthesis of 2,6-Substituted 7-(Het)aryl-7-deazapurine Nucleobases (2,4-Disubstituted 5-(Het)aryl-pyrrolo[2,3-d]pyrimidines)
1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3) Solvents: Acetonitrile, Water; 1 h, 100 °C		By: Sabat, Nazarii; et al Synthesis (2017), 49(20), 4623-4650.
Experimental Protocols		

Scheme 33 (2 Reactions)

Steps: 1 Yield: 98%

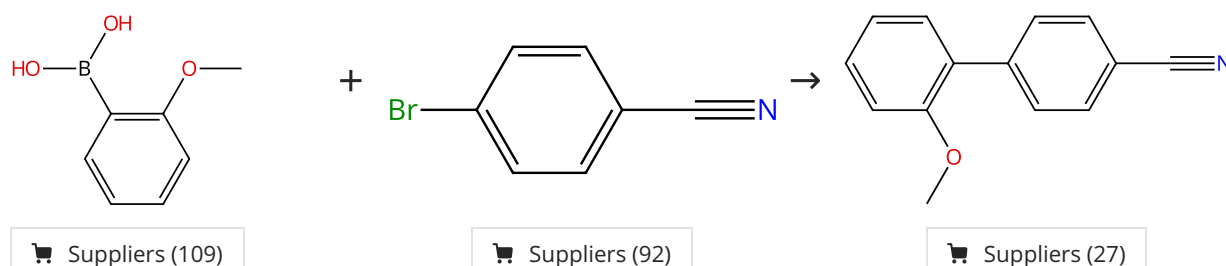


31-179-CAS-23798478	Steps: 1 Yield: 98%	Method for preparing anthracene-containing dimer skeleton ligand, and application thereof in metal-catalyzed reaction
1.1 Reagents: Dipotassium phosphate Catalysts: (<i>SP</i> 4-3)- <i>rel</i> -(Methanesulfonato-κ <i>O</i>)[2'-[(<i>R</i>)-methyl amino-κ <i>M</i>][1,1'-biphenyl]-2-yl-κ <i>C</i>][[(5 <i>R</i> ,12 <i>S</i>)-5,6,11,12-tetrahydro-5,12[1',2']:6,11[1'',2'']-dibenzenodibenzo[<i>a,e</i>]cycloocten-1-yl]diphenylphosphine]palladium Solvents: Methyl ether; 24 h, 80 °C		Assignee: Technical Institute of Physics and Chemistry, Chinese Academy of Sciences China, CN112250708 A 2021-01-22 PatentPak available

31-179-CAS-20661560	Steps: 1 Yield: 98%	A Monophosphine Ligand Derived from Anthracene Photodimer: Synthetic Applications for Palladium-Catalyzed Coupling Reactions
1.1 Reagents: Tripotassium phosphate Catalysts: (<i>SP</i> 4-3)- <i>rel</i> -(Methanesulfonato-κ <i>O</i>)[2'-[(<i>R</i>)-methyl amino-κ <i>M</i>][1,1'-biphenyl]-2-yl-κ <i>C</i>][[(5 <i>R</i> ,12 <i>S</i>)-5,6,11,12-tetrahydro-5,12[1',2']:6,11[1'',2'']-dibenzenodibenzo[<i>a,e</i>]cycloocten-1-yl]diphenylphosphine]palladium Solvents: 1,2-Dimethoxyethane; 24 h, 80 °C		By: Wang, Xin; et al Organic Letters (2019), 21(20), 8158-8163.
Experimental Protocols		

Scheme 34 (1 Reaction)

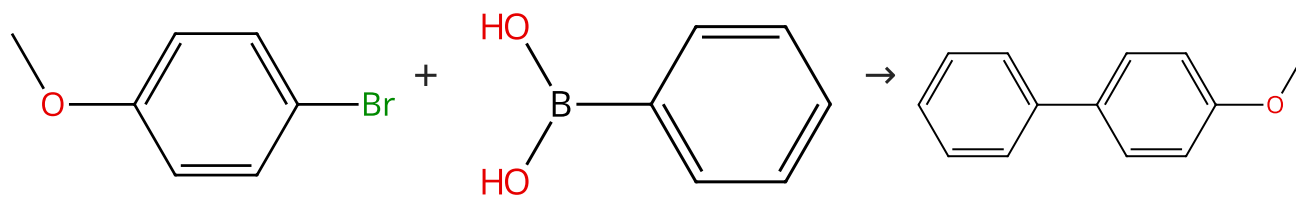
Steps: 1 Yield: 98%



31-614-CAS-39600565	Steps: 1 Yield: 98%	Pyridone palladium catalyst and preparation method and application thereof
1.1 Reagents: Potassium carbonate Catalysts: 3008612-79-8 Solvents: Ethanol; 24 h, 90 °C		Assignee: China Pharmaceutical University China, CN116265117 A 2023-06-20 PatentPak available

Scheme 35 (3 Reactions)

Steps: 1 Yield: 98%



Suppliers (71)

Suppliers (127)

Suppliers (69)

31-179-CAS-10774043

Steps: 1 Yield: 98%

Microfluidic Synthesis of Palladium Nanocrystals Assisted by Supercritical CO₂: Tailored Surface Properties for Applications in Boron Chemistry

By: Gendrineau, Thomas; et al

Angewandte Chemie, International Edition (2012), 51(34), 8525-8528, S8525/1-S8525/23.

1.1 **Reagents:** Potassium carbonate
Catalysts: 2-(Dicyclohexylphosphino)biphenyl (surface functionalized palladium nanocrystals)
Solvents: Toluene; 30 min, 100 °C; 100 °C → rt

Experimental Protocols

31-179-CAS-15037791

Steps: 1 Yield: 98%

Microfluidic Synthesis of Palladium Nanocrystals Assisted by Supercritical CO₂: Tailored Surface Properties for Applications in Boron Chemistry

By: Gendrineau, Thomas; et al

Angewandte Chemie, International Edition (2012), 51(34), 8525-8528, S8525/1-S8525/23.

1.1 **Reagents:** Potassium carbonate
Catalysts: Bis(1,1-dimethylethyl)[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine (surface functionalized palladium nanocrystals)
Solvents: Toluene; 30 min, 100 °C; 100 °C → rt

Experimental Protocols

31-179-CAS-8640620

Steps: 1 Yield: 98%

Microfluidic Synthesis of Palladium Nanocrystals Assisted by Supercritical CO₂: Tailored Surface Properties for Applications in Boron Chemistry

By: Gendrineau, Thomas; et al

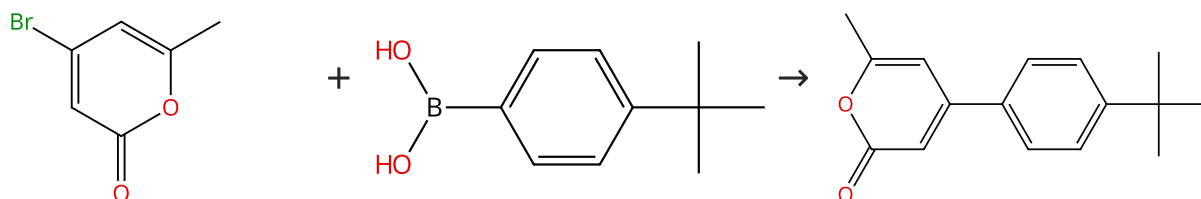
Angewandte Chemie, International Edition (2012), 51(34), 8525-8528, S8525/1-S8525/23.

1.1 **Reagents:** Potassium carbonate
Catalysts: 1,3-Dihydro-1,3-bis(2,4,6-trimethylphenyl)-2*H*-imidazol-2-ylidene (surface functionalized palladium nanocrystals)
Solvents: Toluene; 30 min, 100 °C; 100 °C → rt

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (47)

Suppliers (92)

Supplier (1)

31-179-CAS-16746537

Steps: 1 Yield: 97%

New water soluble Pd-imidate complexes as highly efficient catalysts for the synthesis of C5-arylated pyrimidine nucleosides

By: Kapdi, Anant; et al

RSC Advances (2014), 4(34), 17567-17572.

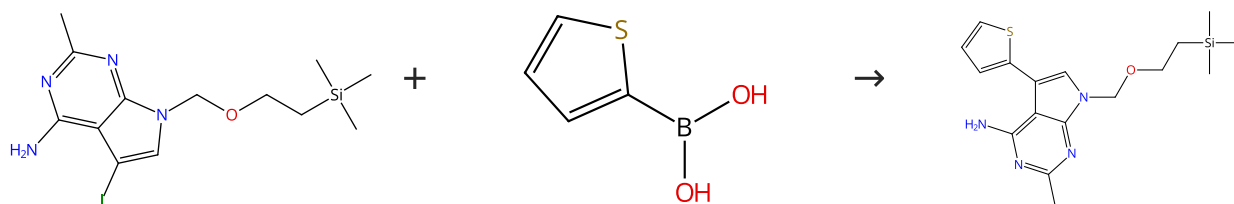
1.1 **Catalysts:** (SP-4-1)-Bis[(1,2-benzisothiazol-3(2*H*)-one-κ^{N2}) 1,1-dioxidato]bis(1,3,5-triaza-7-phosphatricyclo[3.3.1.1^{3,7}]decane-κ^{P7})palladium
Solvents: Water; 5 min, rt; 5 min, rt

1.2 **Reagents:** Sodium carbonate; 0.5 h, 50 °C

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (102)

31-179-CAS-18370398

Steps: 1 Yield: 97%

Synthesis of 2,6-Substituted 7-(Het)aryl-7-deazapurine Nucleobases (2,4-Disubstituted 5-(Het)aryl-pyrrolo[2,3-d]pyrimidines)

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Palladium diacetate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3)
Solvents: Acetonitrile, Water; 3 h, 100 °C

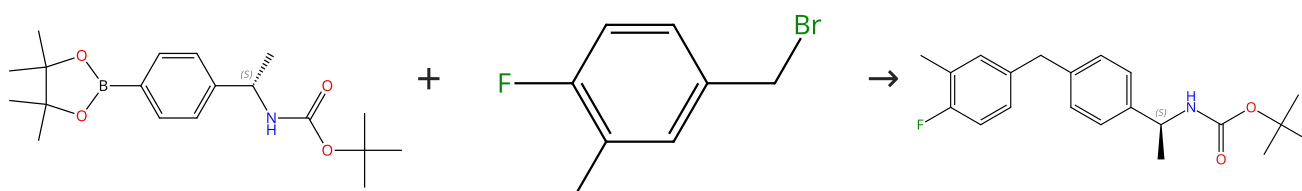
By: Sabat, Nazarii; et al

Synthesis (2017), 49(20), 4623-4650.

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 97%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (47)

Suppliers (49)

31-614-CAS-43011156

Steps: 1 Yield: 97%

Preparation of benzenesulfonamide compounds for the treatment of skin depigmentation or depigmentation-related diseases

- 1.1 **Reagents:** Potassium carbonate
Solvents: Acetonitrile, Water; rt → reflux; 12 h, reflux

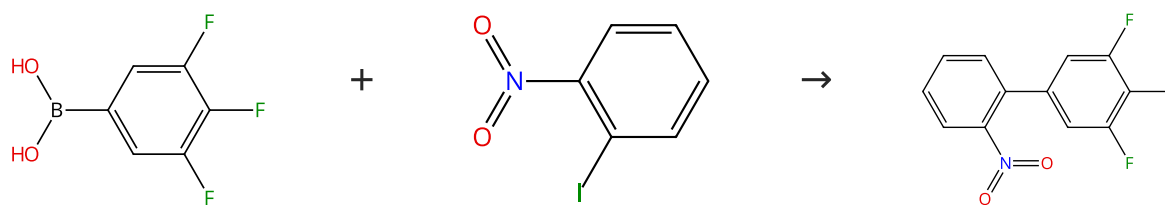
Assignee: China Pharmaceutical University

China, CN118108676 A 2024-05-31

PatentPak available

Scheme 39 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (93)

Suppliers (67)

Suppliers (29)

31-614-CAS-39600661

Steps: 1 Yield: 97%

Pyridone palladium catalyst and preparation method and application thereof

- 1.1 **Reagents:** Potassium carbonate
Catalysts: 3008612-79-8
Solvents: Ethanol, Water; 24 h, 90 °C

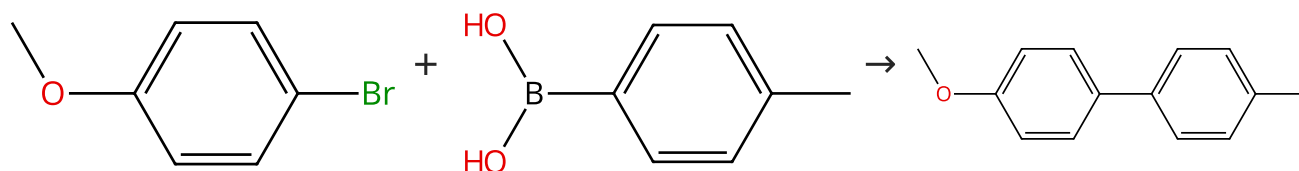
Assignee: China Pharmaceutical University

China, CN116265117 A 2023-06-20

PatentPak available

Scheme 40 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (71)

Suppliers (114)

Suppliers (43)

31-179-CAS-16746525

Steps: 1 Yield: 97%

1.1 **Catalysts:** (*SP*-4-1)-Bis[(1,2-benzisothiazol-3(2*H*)-one- κN^2) 1,1-dioxidato]bis(1,3,5-triaza-7-phosphatricyclo[3.3.1.1^{3,7}]decane- κP^7)palladium

Solvents: Water; 5 min, rt; 5 min, rt

1.2 **Reagents:** Sodium carbonate; 0.5 h, 50 °C

Experimental Protocols

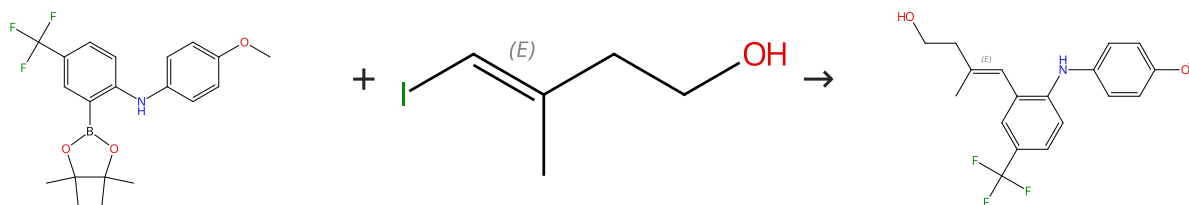
New water soluble Pd-imidate complexes as highly efficient catalysts for the synthesis of C5-arylated pyrimidine nucleosides

By: Kapdi, Anant; et al

RSC Advances (2014), 4(34), 17567-17572.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 96%



Supplier (1)

Double bond geometry shown

Suppliers (4)

Double bond geometry shown

31-179-CAS-4817940

Steps: 1 Yield: 96%

1.1 **Reagents:** Tripotassium phosphate

Catalysts: Palladium diacetate, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine

Solvents: Ethanol, Tetrahydrofuran, Water; 24 h, 90 °C

Experimental Protocols

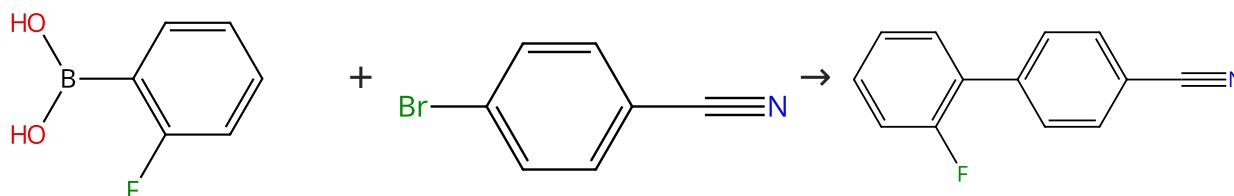
Diastereoselective Oxidative C-N/C-O and C-N/C-N Bond Formation Tandems Initiated by Visible Light: Synthesis of Fused N-Arylindolines

By: Morris, Scott A.; et al

Advanced Synthesis & Catalysis (2015), 357(10), 2311-2316.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (113)

Suppliers (92)

Suppliers (35)

31-614-CAS-39600573

Steps: 1 Yield: 96%

1.1 **Reagents:** Potassium carbonate

Catalysts: 3008612-79-8

Solvents: Ethanol; 24 h, 90 °C

Pyridone palladium catalyst and preparation method and application thereof

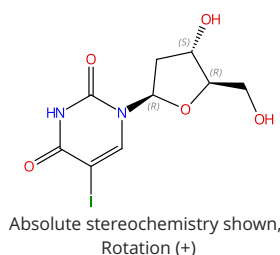
Assignee: China Pharmaceutical University

China, CN116265117 A 2023-06-20

PatentPak available

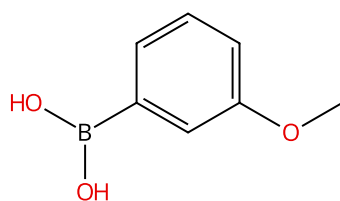
Scheme 43 (1 Reaction)

Steps: 1 Yield: 96%



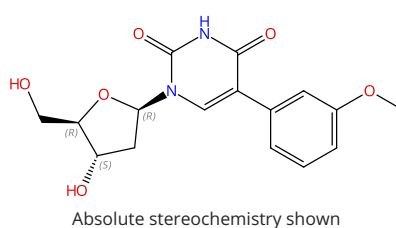
Suppliers (109)

+



Suppliers (109)

→



Supplier (1)

31-179-CAS-16746513

Steps: 1 Yield: 96%

1.1 **Catalysts:** (*SP*-4-1)-Bis(1*H*-isoindole-1,3(2*H*)-dionato-κ*N*²)bis(1,3,5-triaza-7-phosphatricyclo[3.3.1.1^{3,7}]decane-κ*P*⁷)palladium
Solvents: Water; 5 min, rt; 5 min, 80 °C

1.2 **Reagents:** Triethylamine
Solvents: Water; 6 h, 80 °C

Experimental Protocols

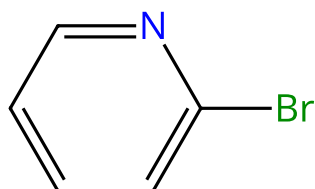
New water soluble Pd-imidate complexes as highly efficient catalysts for the synthesis of C5-arylated pyrimidine nucleosides

By: Kapdi, Anant; et al

RSC Advances (2014), 4(34), 17567-17572.

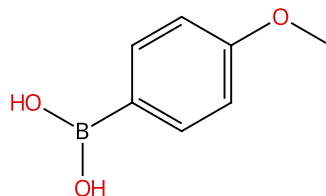
Scheme 44 (1 Reaction)

Steps: 1 Yield: 96%



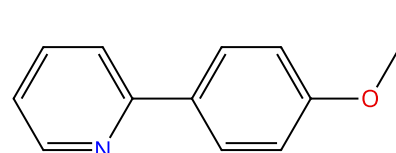
Suppliers (87)

+



Suppliers (113)

→



Suppliers (64)

31-179-CAS-16746532

Steps: 1 Yield: 96%

1.1 **Catalysts:** (*SP*-4-1)-Bis[(1,2-benzisothiazol-3(2*H*)-one-κ*N*²) 1,1-dioxidato]bis(1,3,5-triaza-7-phosphatricyclo[3.3.1.1^{3,7}]decane-κ*P*⁷)palladium
Solvents: Water; 5 min, rt; 5 min, rt

1.2 **Reagents:** Sodium carbonate; 0.5 h, 50 °C

Experimental Protocols

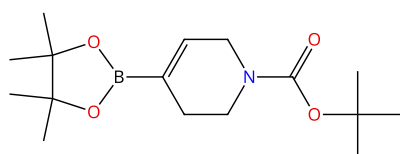
New water soluble Pd-imidate complexes as highly efficient catalysts for the synthesis of C5-arylated pyrimidine nucleosides

By: Kapdi, Anant; et al

RSC Advances (2014), 4(34), 17567-17572.

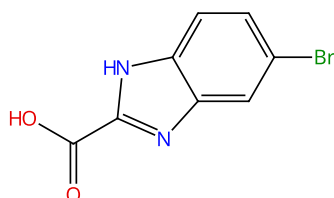
Scheme 45 (1 Reaction)

Steps: 1 Yield: 96%



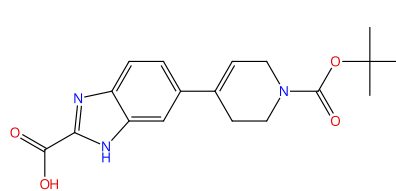
Suppliers (120)

+



Suppliers (71)

→



31-614-CAS-31412551

Steps: 1 Yield: 96%

1.1 **Reagents:** Sodium carbonate
Catalysts: (*SP*-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino-κ*P*]ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; overnight, 95 °C

Development of a Practical and Scalable Synthetic Route for the Adenosine Monophosphate-Activated Protein Kinase Activator, ASP4132

By: Tajima, Takayoshi; et al

Organic Process Research & Development (2021), 25(12), 2786-2794.

Copyright © 2025 American Chemical Society (ACS). All Rights Reserved.

Internal use only. Redistribution is subject to the terms of your CAS SciFinder License Agreement and CAS information Use Policies.